LECTURE 22

Logistic Regression I

Moving from regression to classification.

Goals for this Lecture

Moving away from linear regression – it's time for a new type of model

- Introduce a new task: classification
- Deriving a new model to handle this task

Agenda

- Regression vs. Classification
- The Logistic Regression Model
- Cross-Entropy Loss

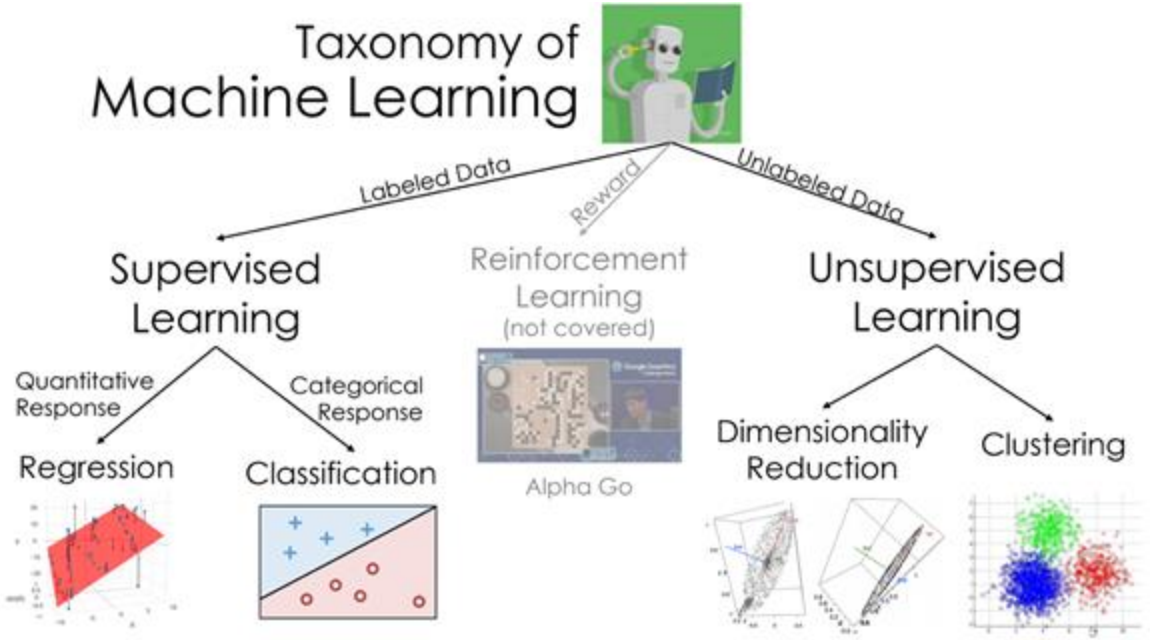
Regression vs Classification

- **Regression vs. Classification**
- The Logistic Regression Model
- Cross-Entropy Loss

Beyond Regression

Up until this point, we have been working exclusively with linear regression.

In the machine learning “landscape,” there are many other types of models.



Beyond Regression

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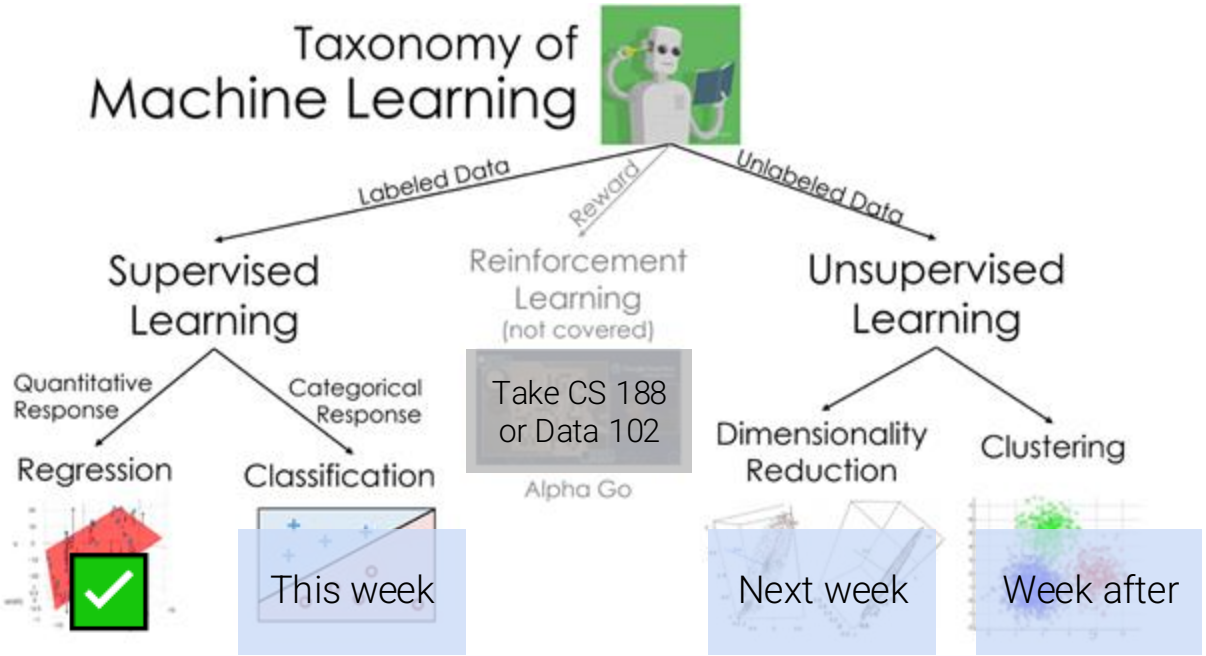
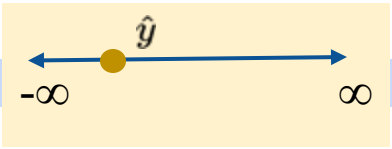


So Far: Regression

In regression, we use unbounded numeric features to predict an *unbounded numeric output*.

GAME_ID	TEAM_NAME	MATCHUP	REB	FTM	TOV	GOAL_DIFF	WON
21700001	Boston Celtics	BOS @ CLE	46	19	12	-0.049	0
21700002	Golden State Warriors	GSW vs. HOU	41	19	17	0.053	0
21700003	Charlotte Hornets	CHA @ DET	47	23	17	-0.030	0
21700004	Indiana Pacers	IND vs. BKN	47	25	14	0.041	1
21700005	Orlando Magic	ORL vs. MIA	50	22	15	0.042	1

$x^\top \theta$



Input: numeric features

Model: linear combination

Output: numeric prediction

Examples:

- Predict goal difference from turnover %
- Predict tip from total bill
- Predict mpg from hp

Now: Classification

In **classification**, we use unbounded numeric features to predict a *categorical class*.

Examples:

- Predict *which team won* from turnover %
- Predict *day of week* from total bill
- Predict *model of car* from hp

GAME_ID	TEAM_NAME	MATCHUP	REB	FTM	TOV	GOAL_DIFF	WON
21700001	Boston Celtics	BOS @ CLE	46	19	12	-0.049	0
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Input: numeric features

$$p = \sigma(x^\top \theta)$$

Model: linear combination transformed by non-linear **sigmoid**

Win?
If p > 0.5: predict a win
Other: predict a loss

Decision rule

Output: **class**

An aside: we will use logistic "regression" to perform a *classification* task. Here, "regression" refers to the type of model, not the task being performed.

Kinds of Classification

We are interested in predicting some **categorical variable**, or **response**, y.

Binary classification

- Two classes
- **Responses** y are either 0 or 1

win or lose

disease or no disease

spam or ham

Multiclass classification

- Many classes
- Examples: Image labeling (Pishi, Thor, Hera), next word in a sentence, etc.

Structured prediction tasks

- Multiple related classification predictions
- Examples: Translation, voice recognition, etc.

Our new goal: predict a **binary** output (y_hat = 0 or y_hat = 1) given inputted numeric features

The Modeling Process

	Regression ($y \in \mathbb{R}$)	**Classification** ($y \in \{0, 1\}$)
1. Choose a model	Linear Regression $\hat{y} = f_\theta(x) = x^T\theta$	??
2. Choose a loss function	Squared Loss or Absolute Loss	??
3. Fit the model	Regularization Sklearn/Gradient descent	Regularization Sklearn/Gradient descent
4. Evaluate model performance	R^2, Residuals, etc.	?? (next lecture)

The Logistic Regression Model

- Regression vs. Classification
- **The Logistic Regression Model**
- Cross-Entropy Loss

The `games` Dataset

New modeling task, new dataset.

The `games` dataset describes the win/loss results of basketball teams.

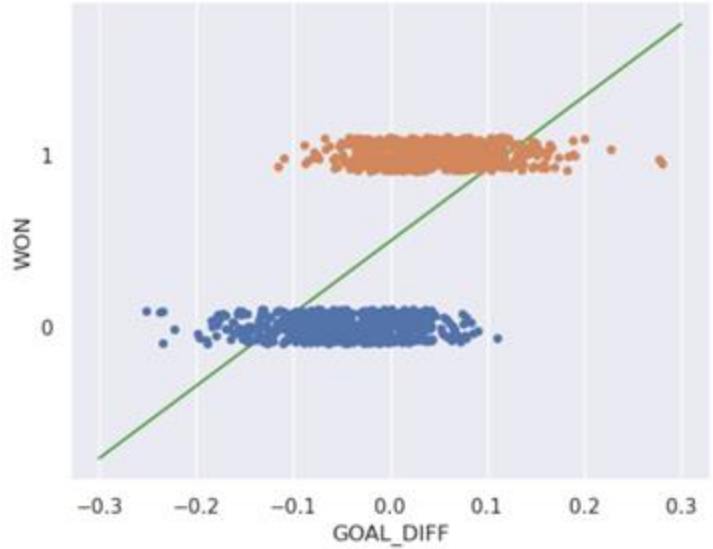
GAME_ID	TEAM_NAME	MATCHUP	WON	GOAL_DIFF
21700001	Boston Celtics	BOS @ CLE	0	-0.049
21700002	Golden State Warriors	GSW vs. HOU	0	0.053
21700003	Charlotte Hornets	CHA @ DET	0	-0.030
21700004	Indiana Pacers	IND vs. BKN	1	0.041
21700005	Orlando Magic	ORL vs. MIA	1	0.042

Difference in field goal success rate between teams

If a team won their game, we say they are in "Class 1"

Why Not Least Squares Linear Regression?

> I suppose it is tempting, if the only tool you have is a hammer, to treat everything as if it were a nail.
>
> – Abraham Maslow, *The Psychology of Science*



Problems:

- The output $\hat{y}$ can be outside the label range {0, 1}.
- Some outputs can't be interpreted: what does a class of "-2.3" mean?

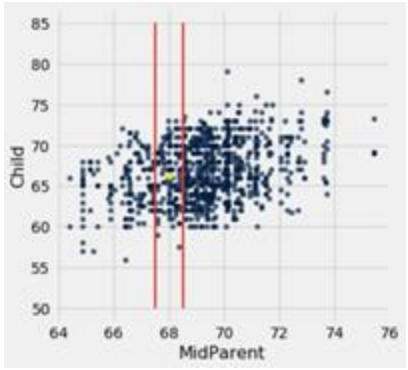
Demo

Back to Basics: the Graph of Averages

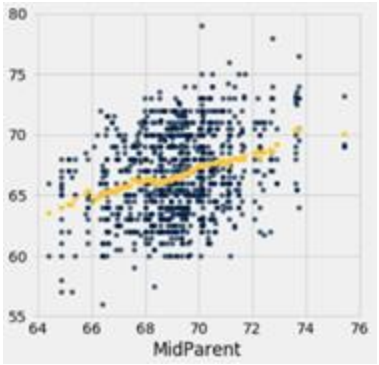
Clearly, least squares regression won't work here. We need to start fresh.

In Data 8 book, you built up to the idea of linear regression by considering the **graph of averages**.

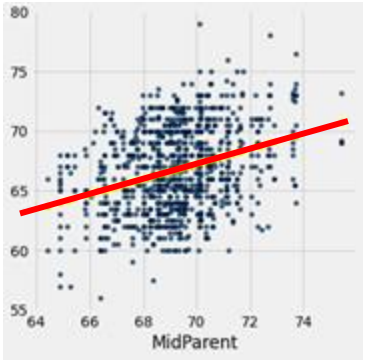
> For an input x, compute the **average value of y for all nearby x**, and predict that.
>
> [Data 8 textbook]



Bucket x-axis into bins



Average y vs. x bins is approximately linear



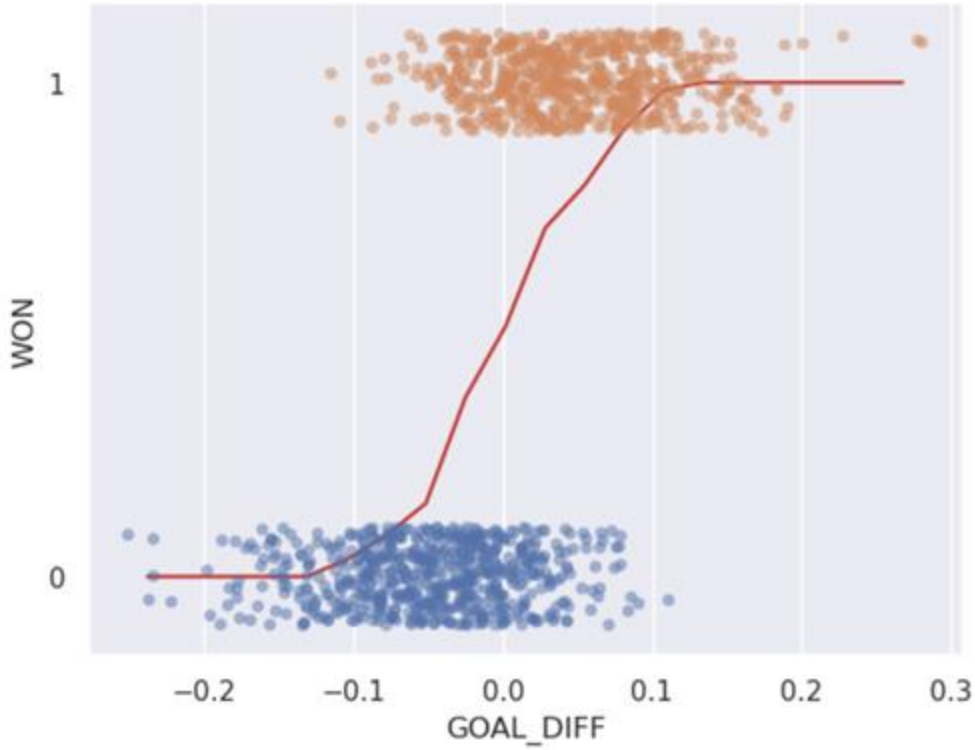
Parametric linear model

$$\hat{y} = x^T\theta$$

Graph of Averages for Our Classification Task

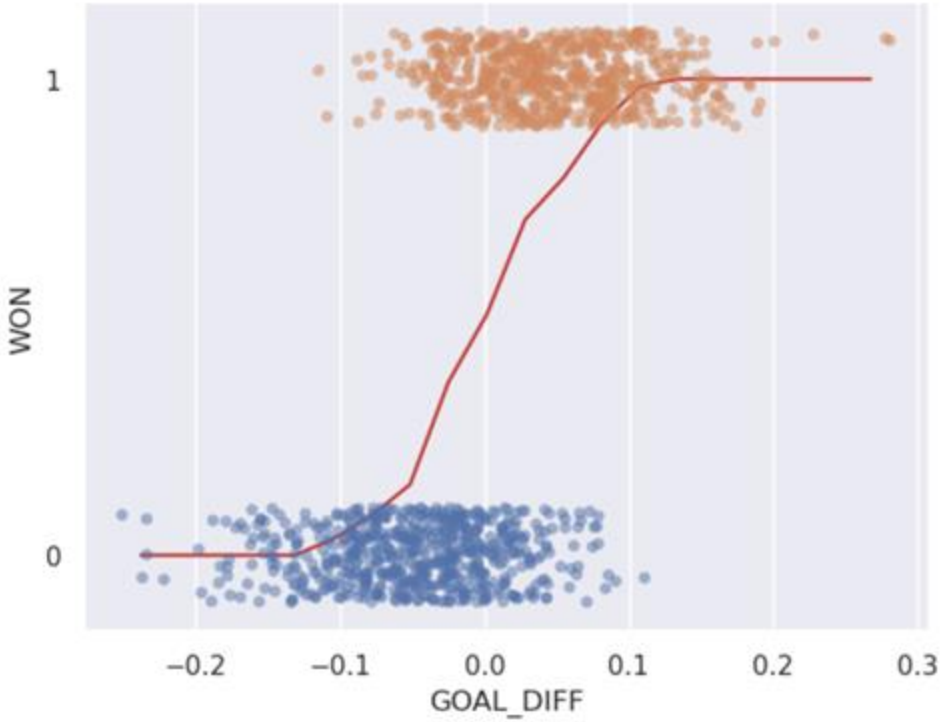
> For an input x, compute the **average value of y for all nearby x**, and predict that.

[textbook]



Demo

Graph of Averages for Our Classification Task



Looking a lot better!

Some observations:

- All predictions are between 0 and 1.
- Our predictions are **non-linear.**
  - Specifically, we see an “S” shape.

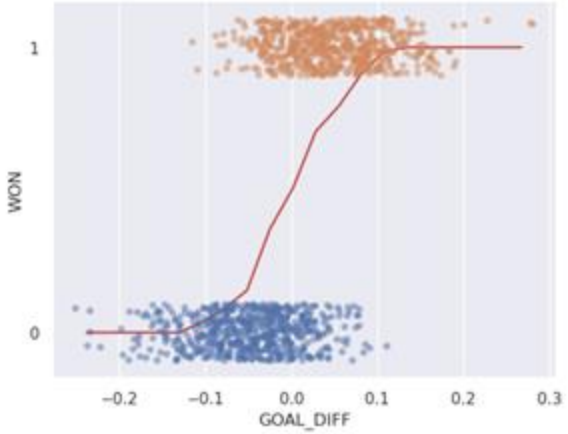
    This will be important soon.

Graph of Averages for Our Classification Task

Thinking more deeply about what we've just done.

To compute the average of a bin, we:

- Counted the number of wins in the bin.
- Divided by the total number of datapoints in the bin.



$$\frac{1(\#\, Y{=}1 \text{ in bin}) + 0(\#\, Y{=}0 \text{ in bin})}{\#\text{ datapoints in bin}} = \frac{\#\, Y{=}1 \text{ in bin}}{\#\text{ datapoints in bin}} = P(Y = 1 \,|\, \text{bin})$$

This is the probability that Y is 1 in that bin!

Predicting Probabilities

Our curve is modeling the *probability* that Y = 1 for a particular value of x.

$$p = P(Y = 1 \mid x)$$

- This matches our observation that all predictions are between 0 and 1 – the predictions are representing probabilities.

New modeling goal: model the *probability* of a datapoint belonging to Class 1.

Handling the Non-Linear Output

New modeling goal: model the *probability* of a datapoint belonging to Class 1.

We still seem to have a problem: our probability curve is non-linear, but we only know linear modeling techniques.

Good news: we've seen this before! To capture non-linear relationships, we:

1. Applied transformations to linearize the relationship.
2. Fit a linear model to the transformed variables.
3. Transformed the variables back to find their underlying relationship.

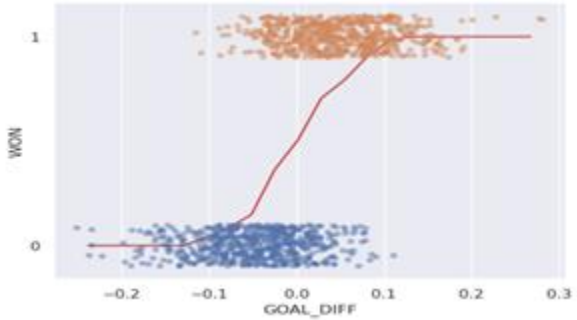
We'll use the exact same process here.

Step 1: Linearize the Relationship

Our S-shaped probability curve doesn't match any relationship we've seen before. The bulge diagram won't help here.

To understand what transformation to apply, think about our eventual goal: assign each datapoint to its *most likely* class.

How do we decide which class is more likely? One way: check which class has the higher predicted probability.

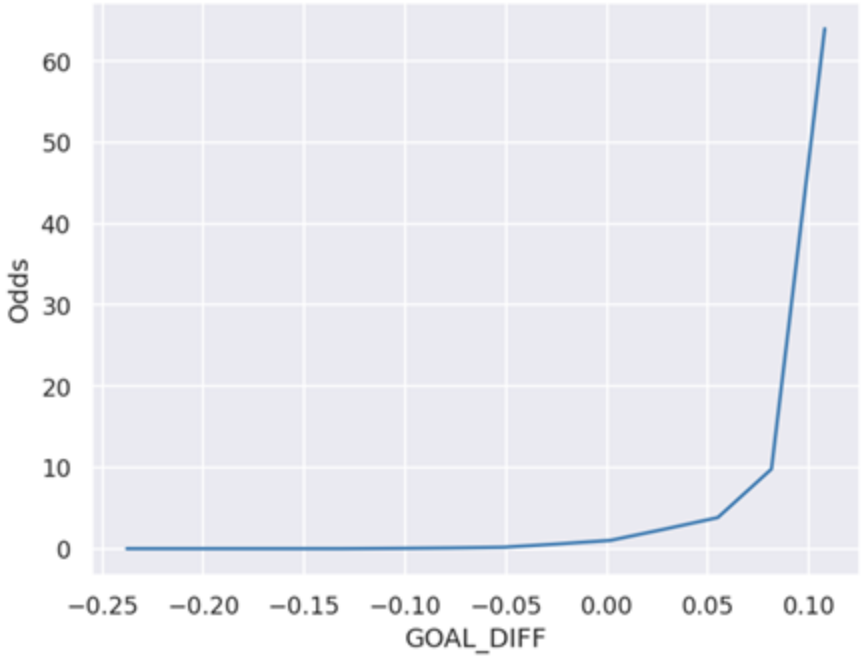


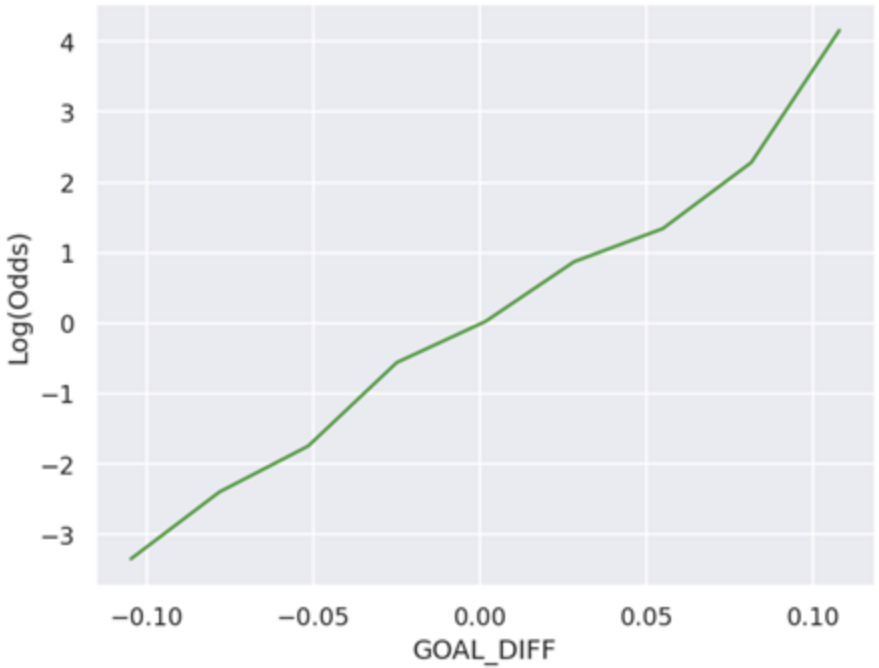
"Odds" is defined as the ratio of the probability of Y being Class 1 to the probability of Y being Class 0

$$\text{odds} = \frac{P(Y=1\,|\,x)}{P(Y=0\,|\,x)} = \frac{p}{1-p}$$

Step 1: Linearize the Relationship

The odds curve looks roughly exponential. To linearize it, we'll take the logarithm*.

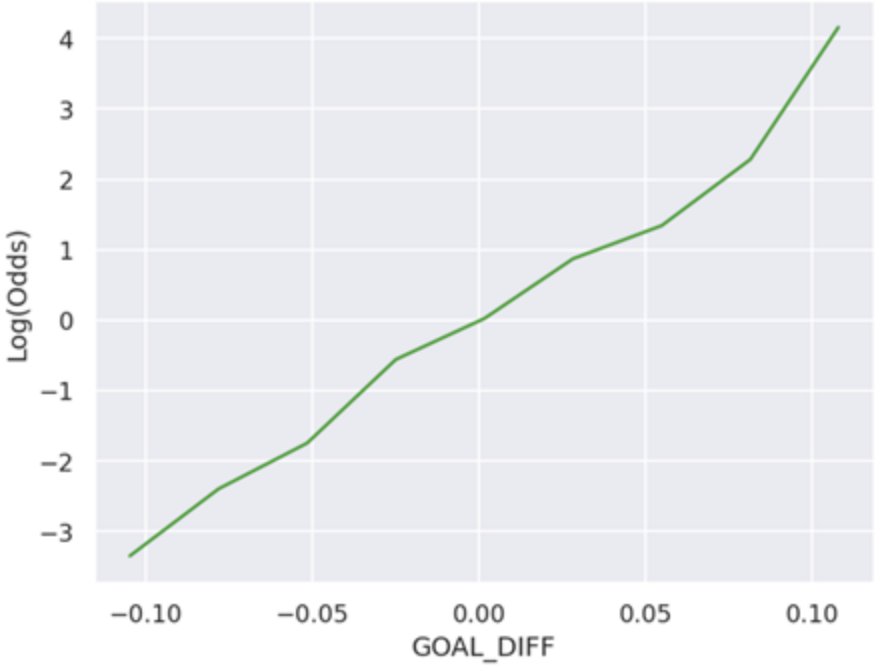




*In Data 100: assume that "log" means base e natural log unless told otherwise.

Step 2: Find a Linear Combination of the Features

We now have a linear relationship between our transformed variables. We can represent this relationship as a linear combination of the features.



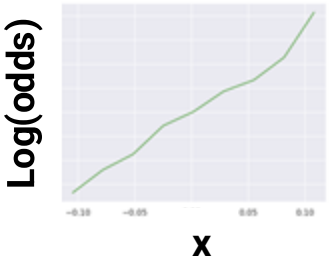
Our linear combination

$$z = x^\top \theta = \theta_0 + \theta_1 x_1 + \ldots + \theta_p x_p$$

Remember that our goal is to predict the *probability* of Class 1. This linear combination isn't our final prediction! We use z instead of y_hat to remind ourselves.

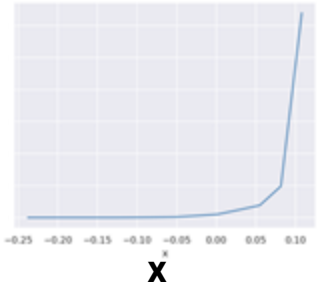
$$z = \log(\text{odds}) = \log\left(\frac{p}{1-p}\right)$$

Step 3: Transform Back to Original Variables

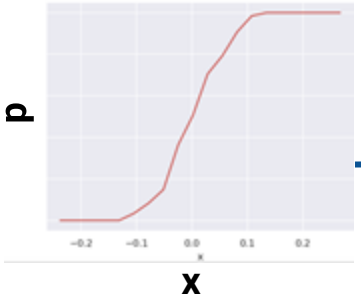


Solve for p:

$$\log\left(\frac{p}{1-p}\right) = x^T\theta$$



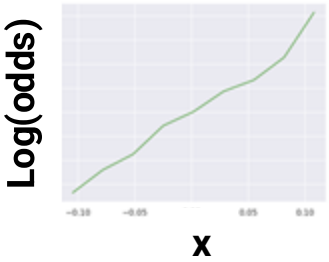
$$\frac{p}{1-p} = e^{x^T\theta}$$



$$p = \frac{1}{1+e^{-x^T\theta}}$$

This is called the **logistic function**, $\sigma(\ )$.

Step 3: Transform Back to Original Variables



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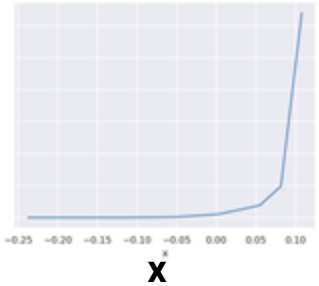
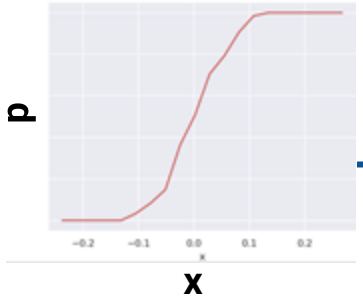


$$\frac{p}{1-p} = e^{x^T\theta}$$

$$p = e^{x^T\theta} - pe^{x^T\theta}$$

$$p = \frac{e^{x^T\theta}}{1+e^{x^T\theta}}$$

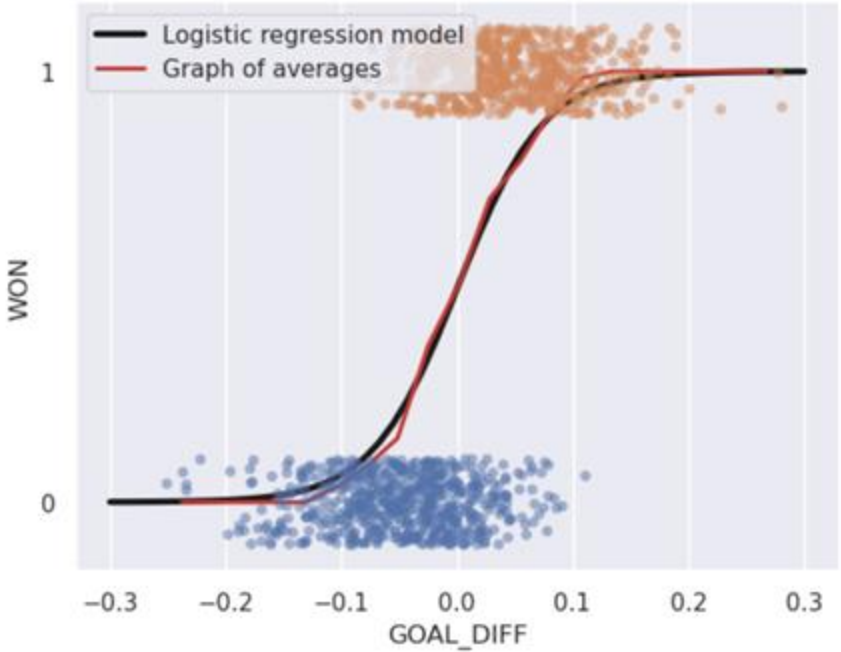


$$p = \frac{1}{1+e^{-x^T\theta}}$$

This is called the **logistic function**, $\sigma(\ )$.

Arriving at the Logistic Regression Model

We have just derived the **logistic regression model** for the probability of a datapoint belonging to Class 1.



$$P(Y=1\,|\,x) = \frac{1}{1+e^{-x^\top\theta}}$$

$$= \frac{1}{1+e^{-(\theta_0+\theta_1 x_1+\ldots+\theta_p x_p)}}$$

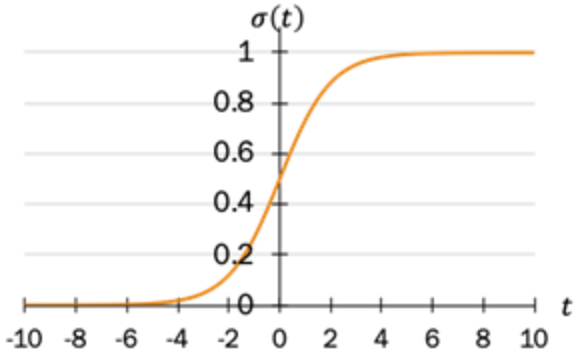
To predict a probability:

- Compute a linear combination of the features, $x^\top\theta$
- Apply the **sigmoid function** $\sigma(x^\top\theta)$

The Sigmoid Function

The S-shaped curve we derived is formally known as the **sigmoid function**.

$$\sigma(t) = \frac{1}{1 + e^{-t}}$$



Reflection/ Symmetry $1 - \sigma(t) = \frac{e^{-t}}{1 + e^{-t}} = \sigma(-t)$

Inverse $t = \sigma^{-1}(p) = \log\left(\frac{p}{1-p}\right)$

Domain $-\infty < t < \infty$

Range $0 < \sigma(t) < 1$

Derivative

$$\frac{d}{dt}\sigma(t) = \sigma(t)(1 - \sigma(t)) = \sigma(t)\sigma(-t)$$

The Sigmoid Converts Numerical Features to Probabilities

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Input: numeric features

$$p = \sigma(x^\top \theta)$$

Model: linear combination transformed by **activation function**

Next lecture:

Win?
If p > 0.5: predict a win
Other: predict a loss

Decision rule

Output: **class**

In logistic regression, the sigmoid transforms a linear combination of numerical features into a **probability.**

$$p = \sigma(x^\top \theta)$$

Formalizing the Logistic Regression Model

Our main takeaways of this section:

- Fit the "S" curve as best as possible.
- The curve models probability: P(Y = 1 | x).
- Assume log-odds is a linear combination of x and θ.

Putting it all together:

$$\hat{P}_\theta(Y = 1|x) = \frac{1}{1 + e^{-x^T\theta}}$$

Estimated probability that given the features x, the response is 1

Logistic function $\sigma(\ )$ at the value $x^T\theta$

The logistic regression model is most commonly written as follows:

$$\hat{P}_\theta(Y = 1|x) = \sigma(x^T\theta)$$

Looks like linear regression. Now wrapped with $\boldsymbol{\sigma}(\ )$!

Example Calculation

Suppose I want to predict the probability that a team wins a game, given **GOAL_DIFF** (first feature) and **number of free throws** (second feature).

I fit a logistic regression model (with no intercept) using my training data, and estimate the optimal parameters:

$$\hat{\theta}^T = [0.1 \quad -0.5]$$

Now, you want to predict the probability that a new team wins their game.

$$x^T = [15 \quad 1]$$

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Suppose x^T contains the GOAL_DIFF and number of free throws for a new team. What is the predicted probability that this new team will win?

Example Calculation

Suppose I want to predict the probability that a team wins a game, given **GOAL_DIFF** (first feature) and **number of free throws** (second feature).

I fit a logistic regression model (with no intercept) using my training data, and estimate the optimal parameters:

$$\hat{\theta}^T = [0.1 \quad -0.5]$$

Now, you want to predict the probability that a new team wins their game.

$$x^T = [15 \quad 1]$$

$$\begin{aligned}\hat{P}_{\hat{\theta}}(Y = 1|x) &= \sigma(x^T\hat{\theta}) \\ &= \sigma(0.1 \cdot 15 + (-0.5) \cdot 1) \\ &= \sigma(1) \\ &= \frac{1}{1 + e^{-1}} \\ &\approx 0.7311\end{aligned}$$

Because the response is more likely to be 1 than 0, a reasonable prediction is

$$\hat{y} = 1$$

(more on this next lecture)

Properties of the Logistic Model

Consider a logistic regression model with one feature and an intercept term (Desmos):

$$p = P(Y = 1 \mid x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

Properties:

- θ_0 controls the position of the curve along the horizontal axis.
- The magnitude of θ_1 controls the "steepness" of the sigmoid.
- The sign of θ_1 controls the orientation of the curve.

Interlude

Classification is hard.

Cross-Entropy Loss

- Regression vs. Classification
- The Logistic Regression Model
- **Cross-Entropy Loss**

The Modeling Process

	Regression ($y \in \mathbb{R}$)	Classification ($y \in \{0, 1\}$)
1. Choose a model	Linear Regression $\hat{y} = f_\theta(x) = x^T\theta$	Logistic Regression $\hat{P}_\theta(Y = 1\|x) = \sigma(x^T\theta)$
2. Choose a loss function	Squared Loss or Absolute Loss	??
3. Fit the model	Regularization Sklearn/Gradient descent	Regularization Sklearn/Gradient descent
4. Evaluate model performance	R^2, Residuals, etc.	?? (next time)

The Modeling Process

	Regression ($y \in \mathbb{R}$)	Classification ($y \in \{0, 1\}$)
1. Choose a model ✓	Linear Regression $\hat{y} = f_\theta(x) = x^T\theta$	Logistic Regression $\hat{P}_\theta(Y = 1\|x) = \sigma(x^T\theta)$
2. Choose a loss function	Squared Loss or Absolute Loss	Can squared loss still work?
3. Fit the model	Regularization Sklearn/Gradient descent	Regularization Sklearn/Gradient descent
4. Evaluate model performance	R^2, Residuals, etc.	?? (next time)

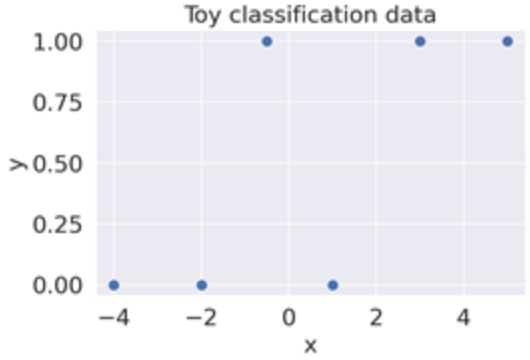
Toy Dataset: L2 Loss

Logistic Regression model:

$$\hat{P}_\theta(Y = 1|x) = \sigma(x^T\theta)$$

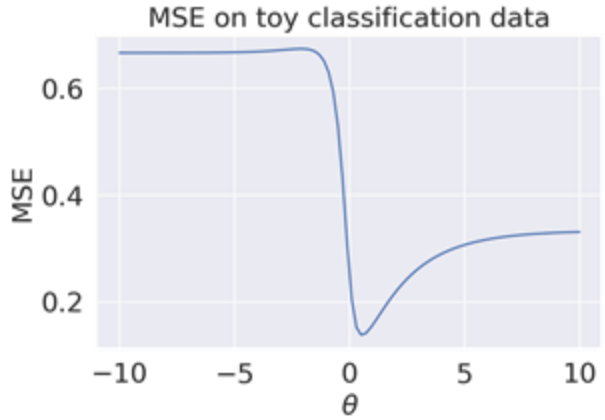
Assume no intercept.
So x, θ both scalars.

	x	y
0	-4.0	0
1	-2.0	0
2	-0.5	1
3	1.0	0
4	3.0	1
5	5.0	1



Mean Squared Error:

$$R(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \sigma(x_i^T\theta)\right)^2$$



The MSE loss surface for logistic regression has many issues!

Demo

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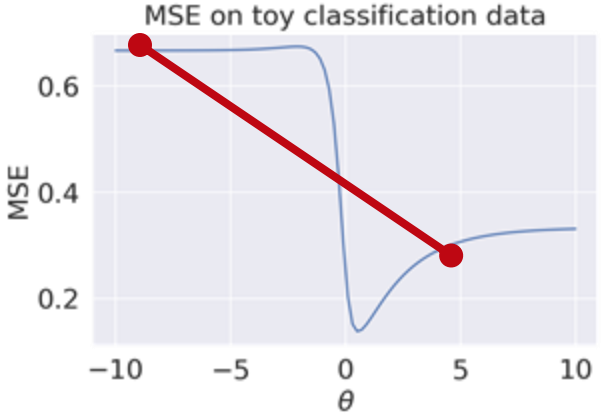
What problems might arise from using MSE loss with logistic regression?

Pitfalls of Squared Loss

$$R(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \sigma(x_i^T\theta)\right)^2$$

1. Non-convex. Gets stuck in local minima.

Secant line crosses function, so $R''(\theta)$ is not greater than 0 for all θ.



Demo

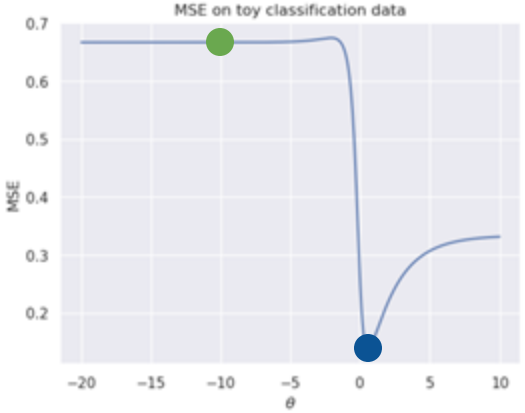
Pitfalls of Squared Loss

$$R(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \sigma(x_i^T\theta)\right)^2$$

1. Non-convex. Gets stuck in local minima.

Secant line crosses function, so $R''(\theta)$ is not greater than 0 for all θ.

Gradient Descent: Different initial guesses will yield different optimal estimates.




```
from scipy.optimize import minimize

minimize(mse_loss_toy_nobias, x0 = 0)["x"][0]
```

```
0.5446601825581691
```

```
minimize(mse_loss_toy_nobias, x0 = -5)["x"][0]
```

```
-10.343653061026611
```

Demo

Pitfalls of Squared Loss

$$R(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \sigma(x_i^T\theta)\right)^2$$

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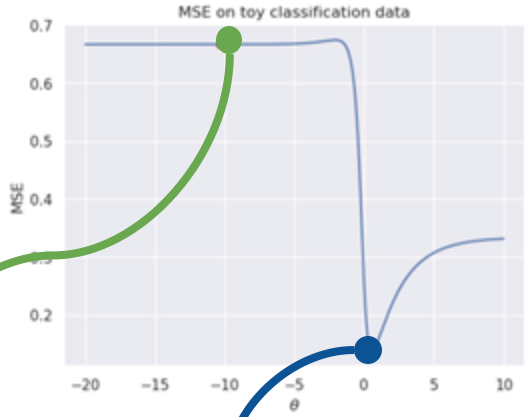
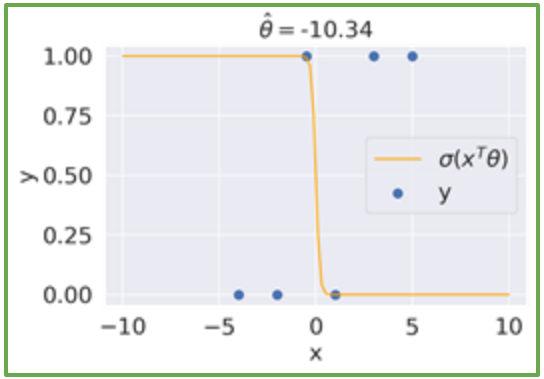
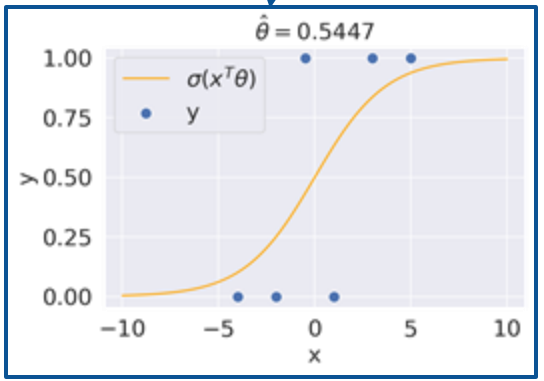






Demo

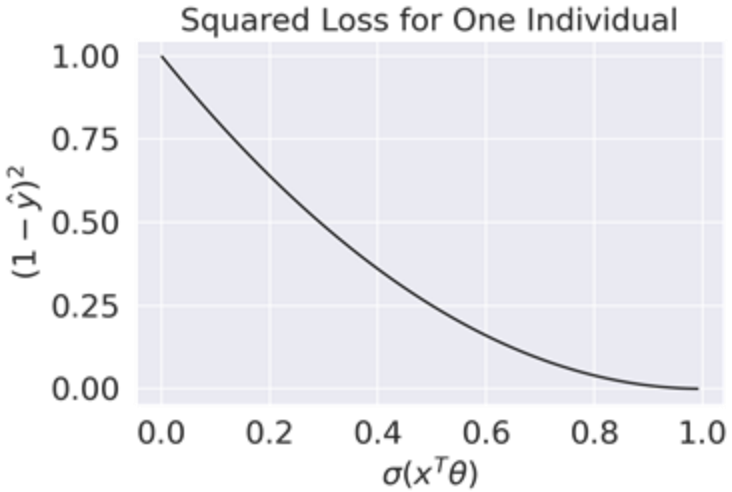
Pitfalls of Squared Loss

$$R(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \sigma(x_i^T\theta)\right)^2$$

1. **Non-convex**. Gets stuck in local minima.

2. **Bounded**. Not a good measure of model error.

- We'd like loss functions to penalize "off" predictions.
- MSE never gets very large, because both response and predicted probability are bounded by 1.



If true y=1 but predicted probability = 0:

$$(y-p)^2 = (1-0)^2 = 1$$

Demo

Choosing a Different Loss Function

	Regression ($y \in \mathbb{R}$)	Classification ($y \in \{0, 1\}$)
1. Choose a model ✓	Linear Regression $\hat{y} = f_\theta(x) = x^T\theta$	Logistic Regression $\hat{P}_\theta(Y = 1\|x) = \sigma(x^T\theta)$
2. Choose a loss function	Squared Loss or Absolute Loss	**Cross-Entropy Loss**
3. Fit the model	Regularization Sklearn/Gradient descent	Regularization Sklearn/Gradient descent
4. Evaluate model performance	R^2, Residuals, etc.	?? (next time)

Loss in Classification

Let y be a binary label {0, 1}, and p be the model's predicted probability of the label being 1.

In a classification task, how do we want our loss function to behave?

- When the true y is 1, we should incur *low* loss when the model predicts large p.
- When the true y is 0, we should incur *high* loss when the model predicts large p.

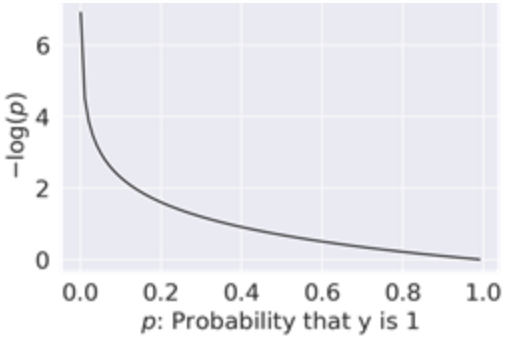
In other words, the behavior we need from our loss function depends on the value of the true class, y.

Cross-Entropy Loss

Let y be a binary label {0, 1}, and p be the probability of the label being 1.

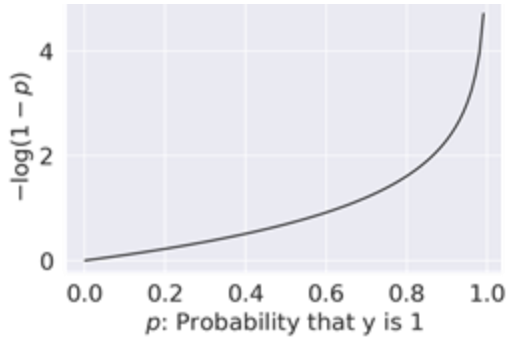
The **cross-entropy loss** is defined as:

$$\text{CE loss} = \begin{cases} -\log(p), & \text{if } y = 1 \\ -\log(1-p), & \text{if } y = 0 \end{cases}$$



For y = 1,

- p → 0: ∞ loss
- p → 1: zero loss



For y = 0,

- p → 0: zero loss
- p → 1: ∞ loss

Cross-Entropy Loss: Two Loss Functions In One!

The piecewise loss function we introduced just then is difficult to optimize – we don't want to check "which" loss to use at each step of optimizing theta.

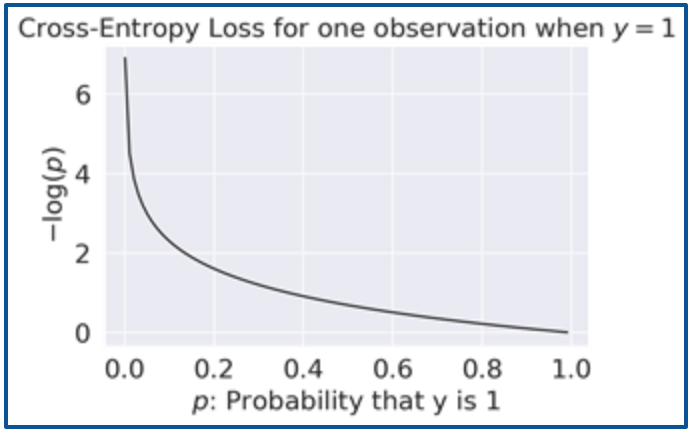
Cross-entropy loss can be equivalently expressed as:

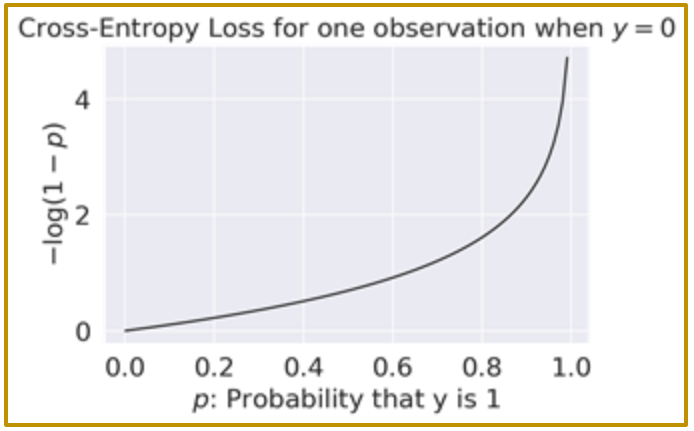
$$\text{CE loss} = \begin{cases} -\log(p), & \text{if } y = 1 \\ -\log(1-p), & \text{if } y = 0 \end{cases} \rightarrow -\left(y \log(p) + (1-y)\log(1-p)\right)$$

makes loss positive

for y = **1**, only this term stays

for y = **0**, only this term stays





Empirical Risk: Average Cross-Entropy Loss

For a single datapoint, the cross-entropy curve is convex. It has a global minimum.

$$-\left(y\log(p)+(1-y)\log(1-p)\right)$$

What about average cross-entropy loss, i.e., empirical risk?

For logistic regression, the empirical risk over a sample of size n is:

$$R(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left(y_i\log(p_i)+(1-y_i)\log(1-p_i)\right)$$

$$= -\frac{1}{n}\sum_{i=1}^{n}\left(y_i\log\left(\sigma\left(X_i^T\theta\right)\right)-(1-y_i)\log\left(1-\sigma\left(X_i^T\theta\right)\right)\right)$$

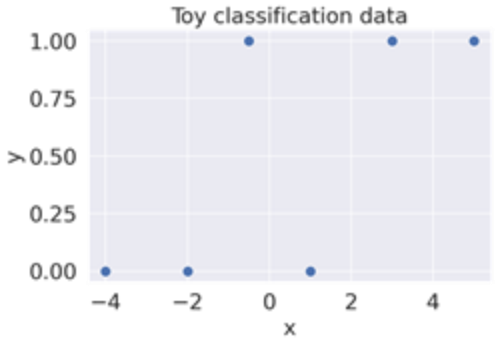
[Recall our model is $\hat{y}_i = p_i = \sigma\left(X_i^T\theta\right)$]

The optimization problem is therefore to find the estimate $\hat{\theta}$ that minimizes R(θ):

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \; -\frac{1}{n}\sum_{i=1}^{n}\left(y_i\log\left(\sigma\left(X_i^T\theta\right)\right)-(1-y_i)\log\left(1-\sigma\left(X_i^T\theta\right)\right)\right)$$

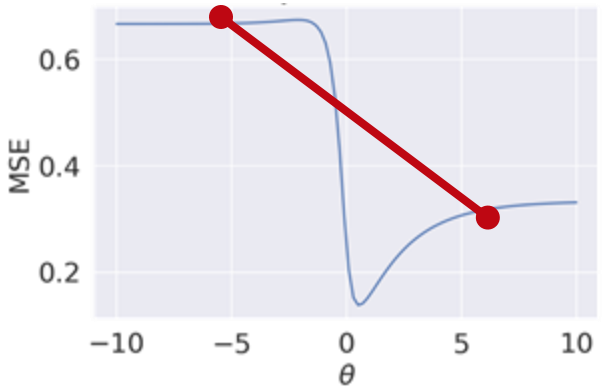
Convexity Proof By Picture

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \; -\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log\left(\sigma\left(X_i^T\theta\right)\right) - (1-y_i)\log\left(1-\sigma\left(X_i^T\theta\right)\right)\right)$$



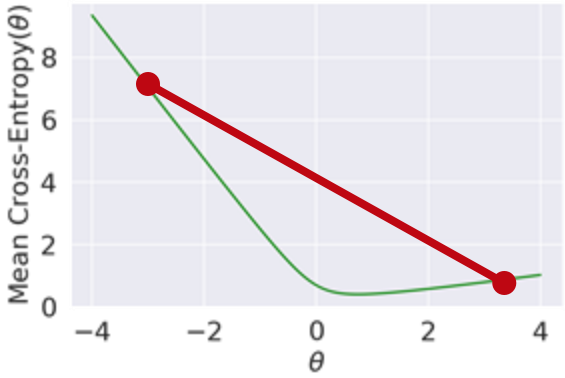
	x	y
0	-4.0	0
1	-2.0	0
2	-0.5	1
3	1.0	0
4	3.0	1
5	5.0	1

Squared Loss Surface



A straight line crosses the curve
Non-convex

Cross-Entropy Loss Surface



Convex!

LECTURE 22

Logistic Regression I

Bonus Material: Maximum Likelihood Estimation

It may have seemed like we just pulled cross-entropy loss out of thin air.

CE loss is justified by a probability analysis technique called maximum likelihood estimation. Read on if you would like to learn more.

Recorded walkthrough: link.

The No-Input Binary Classifier

Suppose you observed some outcomes of a coin (1 = Heads, 0 = Tails):

$$\{0, 0, 1, 1, 1, 1, 0, 0, 0, 0\}$$

Training data has only responses $\mathbb{Y}$ (no features $\mathbb{X}$)

For the next flip, do you predict heads or tails?

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For the next flip, do you predict heads or tails?

A reasonable model is to **assume all flips are IID** (i.e., same coin; same prob. of heads θ).

Parameter θ: Probability that flip == 1 (Heads)

Prediction: 1 or 0

1. Of the below, which is the best theta θ? Why?

A. 0.8 B. 0.5 C. 0.4
D. 0.2 E. Something else

2. For the next flip, would you predict 1 or 0?

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0.4 is the most "intuitive" for two reasons:

1. Frequency of heads in our data.
2. Maximizes the **likelihood** of our data.

The No-Input Binary Classifier

Suppose you observed some outcomes of a coin (1 = Heads, 0 = Tails):

$$\{0, 0, 1, 1, 1, 1, 0, 0, 0, 0\}$$

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1. Of the below, which is the best theta θ? Why?

A. 0.8 **B.** 0.5 **C.** **0.4**

D. 0.2 **E.** Something else

2. For the next flip, would you predict 1 or 0?

The most frequent outcome in the sample which is tails.

Likelihood of Data; Definition of Probability

A Bernoulli random variable Y with parameter p has distribution: $P(Y = y) = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{if } y = 0 \end{cases}$

Given that all flips are IID from the same coin (probability of heads = p), the **likelihood** of our data is **proportional to** the probability of observing the datapoints.

Training data: [0, 0, 1, 1, 1, 1, 0, 0, 0, 0]

Data likelihood: $p^4(1-p)^6$.

Likelihood of Data

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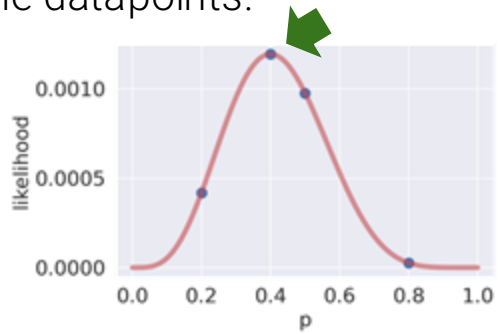

An example of a bad estimate is parameter $p = 0.1$ since the likelihood of observing the training data is going to be :

$$(0.1)^4(0.9)^6 = 0.000053$$

An example of a bad estimate is parameter $p = 0.4$ since the likelihood of observing the training data is going to be :

$$(0.4)^4(0.6)^6 = 0.001194$$

Generalization of the Coin Demo

For training data: {0, 0, 1, 1, 1, 1, 0, 0, 0, 0}

0.4 is the most "intuitive" θ for two reasons:

1. Frequency of heads in our data
2. Maximizes the **likelihood** of our data:

$$\hat{\theta} = \underset{\theta}{\text{argmax}} \left(\theta^4 (1-\theta)^6\right)$$

Parameter θ:
Probability that
IID flip == 1 (Heads)

Prediction:
1 or 0

How can we generalize this notion of likelihood to **any** random binary sample?

$$\{y_1, y_2, \ldots, y_n\} \Rightarrow \hat{\theta} = \underset{\theta}{\text{argmax}} \left(???\right)$$

data (1's and 0's)

likelihood

A Compact Representation of the Bernoulli Probability Distribution

How can we generalize this notion of likelihood to **any** random binary sample?

$$\{y_1, y_2, \ldots, y_n\} \Rightarrow \hat{\theta} = \underset{\theta}{\text{argmax}}\,(???)$$

data (1's and 0's) likelihood

Let Y be Bernoulli(p). The probability distribution can be written compactly:

$$P(Y = y) = p^y (1-p)^{1-y}$$

For P(Y = **1**), only this term stays

For P(Y = **0**), only this term stays

(long, non-compact form):

$$P(Y = y) = \begin{cases} p & \text{if } y = 1 \\ 1-p & \text{if } y = 0 \end{cases}$$

Generalized Likelihood of Binary Data

How can we generalize this notion of likelihood to **any** random binary sample?

$$\{y_1, y_2, \ldots, y_n\} \Rightarrow \hat{\theta} = \underset{\theta}{\text{argmax}}\,(???)$$

data (1's and 0's) — likelihood

Let Y be Bernoulli(p). The probability distribution can be written compactly:

$$P(Y = y) = p^y(1-p)^{1-y}$$

For P(Y = **1**), only this term stays

For P(Y = **0**), only this term stays

If binary data are **IID with same** probability p, then the likelihood of the data is:

$$\prod_{i=1}^{n} p^{y_i}(1-p)^{(1-y_i)}$$

Ex: {0, 0, 1, 1, 1, 1, 0, 0, 0, 0} → $p^4(1-p)^6$

Generalized Likelihood of Binary Data

How can we generalize this notion of likelihood to any random binary sample?

$$\{y_1, y_2, \ldots, y_n\} \Rightarrow \hat{\theta} = \underset{\theta}{\operatorname{argmax}}\,(???)$$

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$$P(Y = y) = p^y (1-p)^{1-y}$$

For P(Y = **1**), only this term stays

For P(Y = **0**), only this term stays

If binary data are **IID with same** probability p, then the likelihood of the data is:

$$\prod_{i=1}^{n} p^{y_i} (1-p)^{(1-y_i)}$$

If data are independent with **different** probability p_i, then the likelihood of the data is:

(spoiler: for logistic regression, $p_i = \sigma(X_i^T \theta)$)

$$\prod_{i=1}^{n} p_i^{y_i} (1-p_i)^{(1-y_i)}$$

Maximum Likelihood Estimation (MLE)

Our **maximum likelihood estimation** problem:

- For i = 1, 2, …, n, let Y_i be independent Bernoulli(p_i). Observe data $\{y_1, y_2, \ldots, y_n\}$.
- We'd like to estimate $p_1, p_2, \ldots, p_n$.

Find $\hat{p}_1, \hat{p}_2, \ldots, \hat{p}_n$ that **maximize** $\prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{(1-y_i)}$

Maximum Likelihood Estimation (MLE)

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Find $\hat{p}_1, \hat{p}_2, \ldots, \hat{p}_n$ that **maximize** $\prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{(1-y_i)}$

Equivalent, simplifying optimization problems (since we need to take the first derivative):

$$\underset{p_1, p_2, \ldots, p_n}{\text{maximize}} \quad \log\left(\prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{(1-y_i)}\right)$$

(log is an increasing function. If a > b, then log(a) > log(b).)

$$= \sum_{i=1}^{n} \log\left(p_i^{y_i}(1-p_i)^{(1-y_i)}\right) = \sum_{i=1}^{n} \left(y_i \log(p_i) + (1-y_i)\log(1-p_i)\right)$$

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Our **maximum likelihood estimation** problem:

- For i = 1, 2, …, n, let Y_i be independent Bernoulli(p_i). Observe data $\{y_1, y_2, \ldots, y_n\}$.
- We'd like to estimate $p_1, p_2, \ldots, p_n$.

Find $\hat{p}_1, \hat{p}_2, \ldots, \hat{p}_n$ that **maximize** $\prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{(1-y_i)}$

Equivalent, simplifying optimization problems (since we need to take the first derivative):

maximize $p_1, p_2, \ldots, p_n$ $\log\left(\prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{(1-y_i)}\right)$ (log is an increasing function. If a > b, then log(a) > log(b).)

$$= \sum_{i=1}^{n} \log\left(p_i^{y_i}(1-p_i)^{(1-y_i)}\right) = \sum_{i=1}^{n} \left(y_i \log(p_i) + (1-y_i)\log(1-p_i)\right)$$

minimize $p_1, p_2, \ldots, p_n$ $-\frac{1}{n}\sum_{i=1}^{n} \left(y_i \log(p_i) + (1-y_i)\log(1-p_i)\right)$

Maximizing Likelihood == Minimizing Average Cross-Entropy

maximize $_{p_1, p_2, \ldots, p_n}$ $\prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{(1-y_i)}$

Log is increasing; max/min properties

minimize $_{p_1, p_2, \ldots, p_n}$ $-\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log(p_i) + (1-y_i)\log(1-p_i)\right)$

For logistic regression, let $p_i = \sigma(X_i^T\theta)$

minimize $_{\theta}$ $\underbrace{-\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log(\sigma(X_i^T\theta)) + (1-y_i)\log(1-\sigma(X_i^T\theta))\right)}$

Cross-Entropy Loss!!

Minimizing cross-entropy loss is equivalent to **maximizing the likelihood of the training data**.

- We are choosing the model parameters that are "most likely", given this data.

Assumption: all data drawn **independently** from the same logistic regression model with parameter θ

- It turns out that many of the model + loss combinations we've seen can be motivated using MLE (OLS, Ridge Regression, etc.)
- You will study MLE further in probability and ML classes. But now you know it exists.

We Did it!

	Regression ($y \in \mathbb{R}$)	Classification ($y \in \{0, 1\}$)
1. Choose a model ✓	Linear Regression $\hat{y} = f_\theta(x) = x^T\theta$	Logistic Regression $\hat{P}_\theta(Y = 1\|x) = \sigma(x^T\theta)$
2. Choose a loss function ✓	Squared Loss or Absolute Loss	Average Cross-Entropy Loss $-\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log(\sigma(X_i^T\theta) + (1 - y_i)\log(1 - \sigma(X_i^T\theta))\right)$
3. Fit the model	Regularization Sklearn/Gradient descent	Regularization Sklearn/Gradient descent
4. Evaluate model performance	R^2, Residuals, etc.	?? (next time)